# Integral Cheat Sheet (A4)

## Basic Trigonometric Integrals

$\int \sin x \, dx = -\cos x + C$

$\int \cos x \, dx = \sin x + C$

$\int dx / \cos^2 x = \tan x + C$

$\int dx / \sin^2 x = -\cot x + C$

## Tangent / Cotangent

$\int \tan x \, dx = -\ln|\cos x| + C$

$\int \cot x \, dx = \ln|\sin x| + C$

## Logarithmic & Exponential

$\int dx/x = \ln|x| + C$

$\int e^x \, dx = e^x + C$

$\int e^{kx} \, dx = (1/k)e^{kx} + C$

## Inverse Trigonometric Forms

$\int dx/(a^2+x^2) = (1/a) \arctan(x/a) + C$

$\int dx/\sqrt{a^2-x^2} = \arcsin(x/a) + C$

$\int dx/\sqrt{a^2+x^2} = \ln|x+\sqrt{a^2+x^2}| + C$

## Substitution Rules

$\int f'(x)/f(x) \, dx = \ln|f(x)| + C$

$\int f(x)f'(x) \, dx = \frac{1}{2}f(x)^2 + C$

## Universal Trigonometric Substitution

$t = \tan(x/2)$

$\sin x = 2t/(1+t^2)$, $\cos x = (1-t^2)/(1+t^2)$

$dx = 2/(1+t^2) \, dt$